

AARON JONES

Electronics | Manufacturing | Test | Embedded Systems | Engineering Automation

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PROFESSIONAL SUMMARY

Computer Engineering Technology professional with hands-on experience supporting electronic systems from prototype assembly and board bring-up through component-level debugging, rework, test, documentation, and manufacturing improvement.

Experienced with wireless power and power-electronics hardware, embedded platforms, laboratory instrumentation, technician support, and engineering workflow automation. Combines practical hardware capability with systems thinking to isolate failures, improve repeatability, and connect engineering, production, and operations.

TECHNICAL SKILLS

Electronics & Hardware: PCB assembly, SMT placement, stencil printing, reflow, soldering, hot-air rework, microscope inspection, board bring-up, analog/digital electronics, power electronics, wireless power, resonant circuits, GaN-based systems, sensors and actuators

Test & Validation: Oscilloscopes, vector network analyzers (VNA), digital multimeters, bench power supplies, continuity/resistance testing, voltage-rail verification, waveform analysis, impedance measurement, functional testing, failure analysis, root-cause troubleshooting

Embedded & Interfaces: STM32/NUCLEO, ESP32/ESP32-CAM, Arduino, Raspberry Pi, GPIO, I2C, UART, SPI, PWM, analog input, interrupts, Wi-Fi, Bluetooth/BLE

Software & Automation: Python, C/C++, Bash, Git/GitHub, VS Code, PlatformIO, STM32CubeIDE, MATLAB, Simulink, LabVIEW, NI-VISA, SCPI, Linux, REST APIs, JSON, FastAPI, React/Vite

Engineering Operations: Manufacturing documentation, assembly/rework procedures, test plans, hardware traceability, BOM normalization, PartsBox, Mouser, Digi-Key, requirements definition, workflow mapping, role-based permissions, technician training, production coordination

PROFESSIONAL EXPERIENCE

Yank Technologies, Inc.

Brooklyn, NY

Electronics Technician - Manufacturing, Test & Engineering Support

Feb 2026-Present

- Support the design, development, debugging, manufacturing, testing, documentation, and verification of wireless power and communication systems across electrical hardware, firmware, software, and system interfaces.
- Execute PCB assembly and component-level rework using stencil printing, SMT placement, reflow, soldering, hot-air tools, microscope inspection, component replacement, and post-rework electrical verification.
- Perform board bring-up and structured failure analysis using schematics, DMMs, bench power supplies, oscilloscopes, VNAs, continuity and resistance checks, voltage-rail measurements, and waveform inspection.
- Troubleshoot power-electronics and resonant systems, including switching stages, custom inductors and coils, reflected impedance, resonant-frequency behavior, and capacitor matching networks.
- Develop assembly procedures, rework instructions, visual inspection checklists, pre-power and power-up checks, functional test records, and board-level traceability documentation.
- Coordinate assembly and manufacturing priorities, guide technicians on soldering, rework, inspection, troubleshooting, and documentation, and communicate technical issues across engineering and production.
- Improve engineering workflows by identifying recurring failure causes, clarifying handoffs, standardizing procedures, and converting undocumented knowledge into repeatable manufacturing and test processes.
- Develop and validate early LabVIEW, NI-VISA, and SCPI instrument-control workflows for commanding bench equipment, reading responses, verifying output states, and troubleshooting communication timeouts.

New York City College of Technology, CUNY - CET Department

Brooklyn, NY

College Lab Technician

June 2023-Present

- Support instructional laboratories involving microcontrollers, digital logic, electronics, and embedded systems; prepare hardware and software environments for scheduled coursework.
- Diagnose wiring faults, sensor failures, power issues, and firmware-upload problems on Arduino, ESP32, STM32/NUCLEO, power supplies, and laboratory equipment.
- Maintain and repair lab boards, sensors, test equipment, and development tools while documenting repairs and guiding students in oscilloscope, DMM, GPIO, and embedded-development workflows.

The Jace Solution

New York, NY

IT Deployment Technician

Sept 2024-2026

- Supported enterprise hardware migrations, device provisioning, configuration testing, ticket documentation, and verification of networked-device and remote-access changes.

ENGINEERING AUTOMATION & PRODUCT LEADERSHIP

AutoBOM - Engineering Workflow Platform | *Product Owner & System Architect* | *Active Development*

- Own the product vision and system architecture for an internal platform connecting engineering, production, purchasing, and inventory through a single traceable workflow.
- Defined product requirements, role-based permissions, entity states, audit trails, push-back and exception workflows, supplier responsibilities, data flows, and cross-functional handoffs.
- Led development of a local Phase 0 platform using a Python/FastAPI backend and React/Vite frontend, with integrations or validated connectivity for Mouser, Digi-Key, PartsBox, and Nexar APIs.
- Developed the original Python/pandas proof of concept for BOM import, column normalization, manufacturer/MPN cleanup, build-quantity scaling, overage calculations, and component-management workflows.
- Direct phased architecture for PostgreSQL, SQLAlchemy/Alembic, Azure AD role synchronization, Microsoft Graph, Docker, CI/CD, and local-to-cloud graceful fallback while reviewing designs for usability, accessibility, and workflow completeness.

SELECTED ENGINEERING INITIATIVES & PROJECTS

Manufacturing Documentation & Hardware Traceability | *Work Instructions, Test Records, Process Control*

- Created structured assembly, rework, inspection, pre-power, functional-test, and verification records that translate engineering requirements into technician-ready execution and preserve board history.

Automated Instrument Control & Test Development | *LabVIEW, NI-VISA, SCPI*

- Built and validated early command/query workflows for programmable bench equipment, including output-state control, response parsing, read/write verification, and timeout troubleshooting.

Multi-Camera CCTV & Edge Monitoring System | *Raspberry Pi, ESP32-CAM, Linux, Networking*

- Designed a distributed camera architecture with networked nodes, centralized viewing and processing, remote monitoring, local logging, and event-alert concepts; resolved camera power and connectivity issues.

Smart Proximity Alert System V2 | *ESP32, Ultrasonic Sensing, ESP32-CAM, Telegram*

- Built a real-time proximity system that activates local indicators and remote image notifications; debugged sensor timing, wireless communication, and power-delivery stability.

Weather & Environmental Monitoring Station | *ESP32, DHT11, OLED, LDR, I2C*

- Integrated temperature, light, display, and status outputs with threshold logic; resolved I2C timeouts through pull-up correction and power-noise isolation.

RC Airboat Control System | *Arduino, Motor Driver, Servo, PWM*

- Implemented dual-motor propulsion, servo steering, battery monitoring, and speed-control logic while diagnosing grounding and current-delivery issues.

STM32 / NUCLEO Development Workflow | *NUCLEO-F446RE, STM32CubeIDE, ST-LINK, Git*

- Built and flashed repository-based firmware, validated target connections, and supported hardware/firmware integration through structured version-control and debugging workflows.

Raspberry Pi Network Monitor & Optimizer | *Linux, Pi-hole, ntopng, Bash*

- Configured network visibility, DNS filtering, traffic monitoring, and scripts for device-management and performance analysis; documented setup and operational behavior.

EDUCATION & CERTIFICATION

New York City College of Technology, CUNY

Bachelor of Technology, Computer Engineering Technology - Magna Cum Laude

Associate in Applied Science, Electromechanical Engineering Technology

Certification: TestOut Routing & Switching Pro

Brooklyn, NY

May 2025

June 2023